AN ASSIGNMENT

ON

MULTIPLE LINEAR REGRESSION & ONE-WAY ANOVA

(with a Classification Extension)

BY

2023A7PS0265U Aditya Sharma

Instructor In-Charge: Dr. Maneesha

prepared for assignment 2 fulfillment of the course

MATH F432 APPLIED STATISTICAL METHODS

BITS PILANI, DUBAI CAMPUS

DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

UAE

PART A — Multiple Linear Regression: Advertising Spend and Sales

1.0 Introduction to Dataset and Tasks

The dataset used is advertising.csv (200 observations), containing advertising spend (in thousands of dollars) across three media -- TV, Radio, and Newspaper -- and the resulting Sales (in thousands of units) for a product. You are required to determine the following:

(a) Fit a multiple linear regression model predicting Sales from TV, Radio, and Newspaper spend.

(b) Test overall model significance (F-test) and each predictor's individual significance (t-tests), and report 95% confidence intervals for each coefficient.

(c) Check the regression assumptions: normality of residuals, homoscedasticity, and multicollinearity among predictors.

2.0 Objective and Hypotheses

This analysis investigates which advertising channels are significant predictors of sales, and how much of the variation in sales the three channels jointly explain.

Overall model (F-test):

H0: $\beta(TV) = \beta(Radio) = \beta(Newspaper) = 0$ (none of the predictors explain sales)

Ha: at least one $\beta(j) \neq 0$

Each coefficient (t-test):

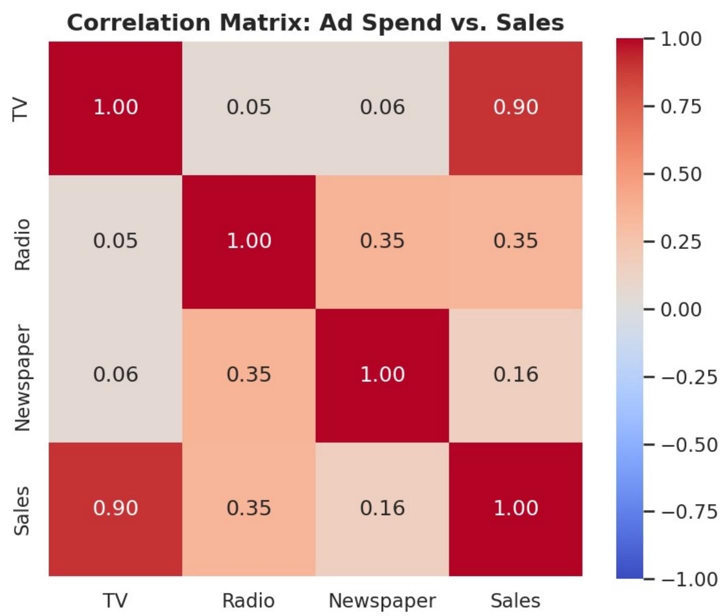
H0: $\beta(j) = 0$ vs. Ha: $\beta(j) \neq 0$, for $j \in \{TV, Radio, Newspaper\}$, at $\alpha = 0.05$.

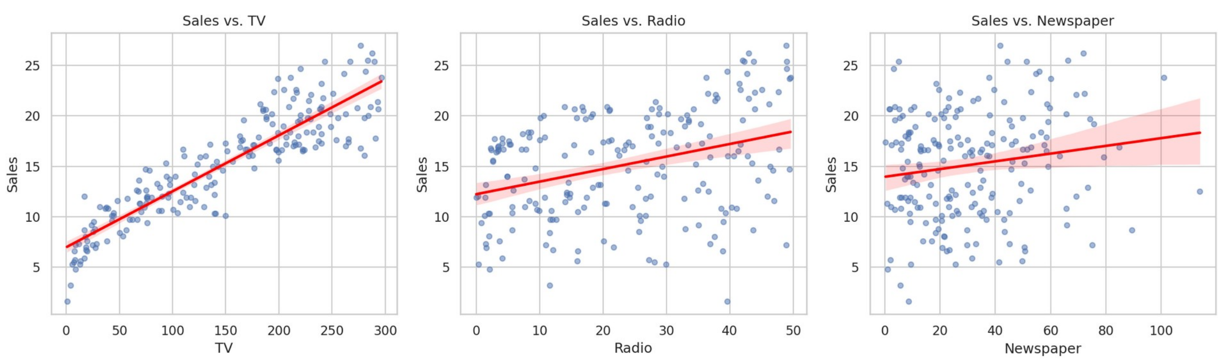
3.0 Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Median	Max
TV	147.04	85.85	0.70	149.75	296.40
Radio	23.26	14.85	0.00	22.90	49.60
Newspaper	30.55	21.78	0.30	25.75	114.00
Sales	15.13	5.28	1.60	16.00	27.00

4.0 Correlation Analysis

TV spend is strongly correlated with Sales ($r = 0.901$); Radio shows a moderate correlation ($r = 0.350$); Newspaper is weakly correlated ($r = 0.158$). Radio and Newspaper are themselves moderately correlated ($r = 0.354$), which is checked for multicollinearity in Section 6.0.





5.0 Regression Results

Fitted model: Sales = 4.6251 + 0.0544 × TV + 0.1070 × Radio + 0.0003 × Newspaper

Predictor	Coefficient	Std. Error	t	p-value	95% CI
Intercept	4.6251	0.308	15.041	< 0.001	(4.019, 5.232)
TV	0.0544	0.001	39.592	< 0.001	(0.052, 0.057)
Radio	0.1070	0.008	12.604	< 0.001	(0.090, 0.124)
Newspaper	0.0003	0.006	0.058	0.954	(-0.011, 0.012)

Overall F-test: $F(3, 196) = 605.4$, $p = 8.13 \times 10^{-99}$ — the model as a whole is highly statistically significant.

$R^2 = 0.903$, Adjusted $R^2 = 0.901$ — the model explains about 90% of the variance in Sales.

Decision on individual predictors: TV and Radio are statistically significant predictors of Sales ($p < 0.001$ each). Newspaper is NOT statistically significant ($p = 0.954$) — its 95% CI for the coefficient, (-0.011, 0.012), contains 0.

6.0 Model Diagnostics

Multicollinearity (Variance Inflation Factor):

Predictor	VIF

TV	2.487
Radio	3.285
Newspaper	3.055

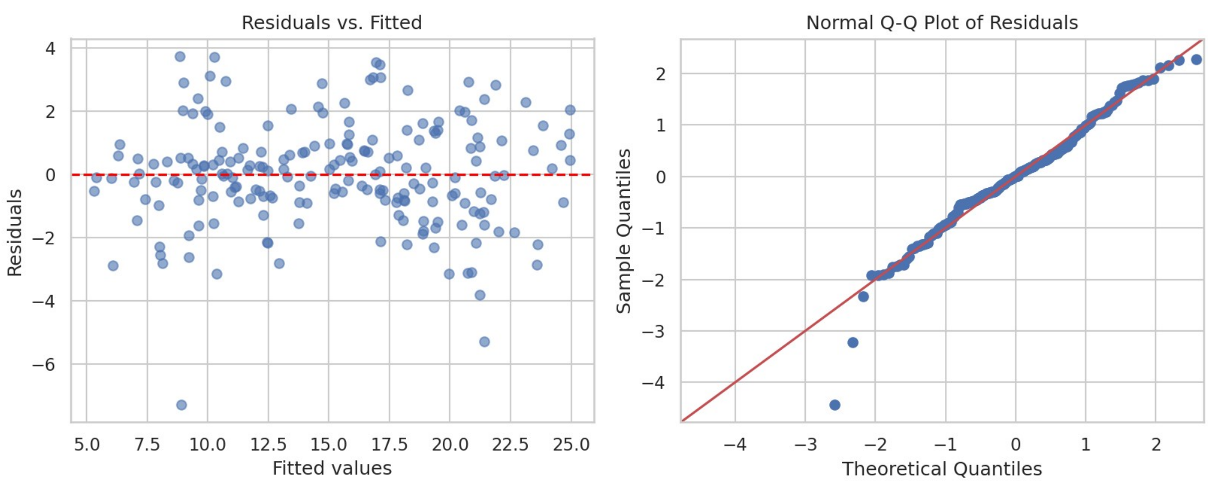
All VIF values are well below the common threshold of 5, so multicollinearity is not a concern.

Normality of residuals (Shapiro-Wilk):

$W = 0.9758$, $p = 0.0016$ — normality is formally rejected at $\alpha = 0.05$ (the residuals are slightly left-skewed with a few heavy-tailed points, skew = -0.431, kurtosis = 4.605), but with $n = 200$ the F- and t-tests remain reasonably robust to mild non-normality by the Central Limit Theorem.

Homoscedasticity (Breusch-Pagan test):

LM statistic = 3.979, $p = 0.264$ — homoscedasticity holds; there is no evidence that residual variance changes with the fitted values.



7.0 Conclusion

TV and Radio advertising spend are statistically significant, positive predictors of Sales; Newspaper spend shows no significant relationship with Sales once TV and Radio are accounted for. The model explains 90.3% of the variance in Sales ($R^2 = 0.903$) and passes the homoscedasticity and multicollinearity checks; residuals show only mild non-normality, which is not a serious concern at this sample size. Practically, this suggests that of the three channels studied, advertising budget is better allocated to TV and Radio than to Newspaper.

8.0 Python Code (Part A)

```
"""
MATH F432 Applied Statistical Methods -- Assignment 2, Part A
Multiple Linear Regression: Sales ~ TV + Radio + Newspaper (advertising.csv)

Tasks:
(a) Fit a multiple linear regression model predicting Sales from TV, Radio,
    and Newspaper advertising spend.
(b) Test overall model significance (F-test) and each coefficient's
    significance (t-tests), report 95% CIs for each coefficient.
(c) Check regression assumptions: normality of residuals, homoscedasticity
    (Breusch-Pagan), and multicollinearity (VIF).
"""
import os
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from scipy import stats
from statsmodels.stats.diagnostic import het_breuschpagan
from statsmodels.stats.outliers_influence import variance_inflation_factor

sns.set(style="whitegrid")

BASE_DIR = os.path.dirname(os.path.abspath(__file__))
FILE_PATH = os.path.join(BASE_DIR, "advertising.csv")
FIG_CORR = os.path.join(BASE_DIR, "fig_correlation_heatmap.png")
FIG_DIAG = os.path.join(BASE_DIR, "fig_regression_diagnostics.png")
FIG_SCATTER = os.path.join(BASE_DIR, "fig_sales_vs_predictors.png")

ALPHA = 0.05

try:
    df = pd.read_csv(FILE_PATH)
except FileNotFoundError:
    print(f"Error: File not found at '{FILE_PATH}'")
    raise SystemExit(1)

predictors = ["TV", "Radio", "Newspaper"]
X = df[predictors]
y = df["Sales"]
n = len(df)

print("--- Descriptive Statistics ---")
print(df[predictors + ["Sales"]].describe().round(2))

print("\n--- Correlation Matrix ---")
corr = df[predictors + ["Sales"]].corr()
print(corr.round(3))

Xc = sm.add_constant(X)
model = sm.OLS(y, Xc).fit()

print("\n--- Multiple Linear Regression: Sales ~ TV + Radio + Newspaper ---")
print(model.summary())

print("\n--- Variance Inflation Factors (multicollinearity check) ---")
vif = pd.DataFrame({
    "feature": predictors,
    "VIF": [variance_inflation_factor(X.values, i) for i in range(X.shape[1])],
})
print(vif.round(3).to_string(index=False))

resid = model.resid
fitted = model.fittedvalues

sh_stat, sh_p = stats.shapiro(resid)
print(f"\nShapiro-Wilk on residuals: W={sh_stat:.4f}, p={sh_p:.4f}")
print("  -> Residuals are", "NOT normal" if sh_p < ALPHA else "approximately normal", f"at 
alpha={ALPHA}")

bp_stat, bp_p, _, _ = het_breuschpagan(resid, Xc)
print(f"Breusch-Pagan test for homoscedasticity: LM stat={bp_stat:.4f}, p={bp_p:.4f}")
print("  ->", "Heteroscedasticity detected" if bp_p < ALPHA else "Homoscedasticity holds", f"at 
alpha={ALPHA}")

print("\n--- Conclusion ---")
```

```
overall_p = model.f_pvalue
print(f"Overall F-test: F={model.fvalue:.1f}, p={overall_p:.3e} "
      f"-> model is {'statistically significant' if overall_p < ALPHA else 'not significant'}")
for pred in predictors:
    p = model.pvalues[pred]
    sig = "significant" if p < ALPHA else "NOT significant"
    print(f"  {pred}: coef={model.params[pred]:.4f}, p={p:.4f} -> {sig}")
print(f"R-squared = {model.rsquared:.3f}  (Adj. R-squared = {model.rsquared_adj:.3f})")

# ---------------------------------------------------------------------------
# Figures
# ---------------------------------------------------------------------------
plt.figure(figsize=(6, 5))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", vmin=-1, vmax=1, square=True)
plt.title("Correlation Matrix: Ad Spend vs. Sales", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.savefig(FIG_CORR, dpi=150)
plt.close()

fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))
for ax, pred in zip(axes, predictors):
    sns.regplot(data=df, x=pred, y="Sales", ax=ax, scatter_kws={"alpha": 0.5, "s": 20},
                line_kws={"color": "red"})
    ax.set_title(f"Sales vs. {pred}")
plt.tight_layout()
plt.savefig(FIG_SCATTER, dpi=150)
plt.close()

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))
axes[0].scatter(fitted, resid, alpha=0.6)
axes[0].axhline(0, color="red", linestyle="--")
axes[0].set_xlabel("Fitted values")
axes[0].set_ylabel("Residuals")
axes[0].set_title("Residuals vs. Fitted")
sm.qqplot(resid, line="45", fit=True, ax=axes[1])
axes[1].set_title("Normal Q-Q Plot of Residuals")
plt.tight_layout()
plt.savefig(FIG_DIAG, dpi=150)
plt.close()

print(f"\nFigures saved: {FIG_CORR}, {FIG_SCATTER}, {FIG_DIAG}")
```

PART B — One-Way ANOVA and Classification: Iris Dataset

9.0 Introduction to Dataset and Tasks

The dataset used is iris.csv (150 observations, Fisher's classic Iris dataset), containing four measurements -- Sepal Length, Sepal Width, Petal Length, and Petal Width, in cm -- for 50 flowers from each of 3 species: setosa, versicolor, and virginica. You are required to determine the following:

(a) Test whether mean Petal Length differs across the three species using a one-way ANOVA.

(b) Check the ANOVA assumptions (normality per group, homogeneity of variance) and apply appropriate post-hoc and robustness checks.

(c) Extension: build a classifier that predicts species from all four measurements, and evaluate it on held-out data.

10.0 Objective and Hypotheses

This analysis investigates whether Petal Length -- the single most discriminative measurement in this dataset -- differs meaningfully across the three iris species.

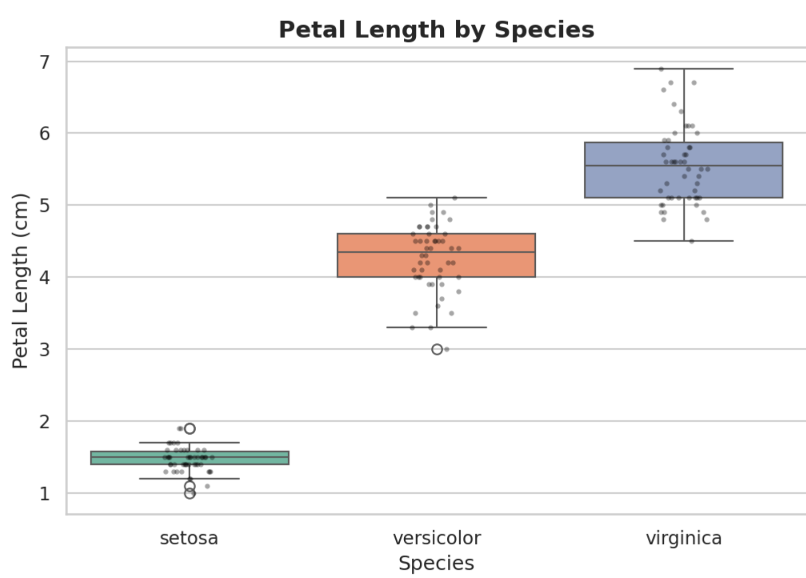
H0: μ(setosa) = μ(versicolor) = μ(virginica)

Ha: at least one species mean differs from the others

Test used: one-way ANOVA (F-test), $\alpha = 0.05$.

11.0 Descriptive Statistics

Species	n	Mean Petal Length	Std. Dev.	Min	Max
setosa	50	1.464	0.174	1.0	1.9
versicolor	50	4.260	0.470	3.0	5.1
virginica	50	5.552	0.552	4.5	6.9



12.0 Assumption Checks

Check	Statistic	p-value	Conclusion
Shapiro-Wilk (setosa)	W = 0.9549	0.0547	approx. normal
Shapiro-Wilk (versicolor)	W = 0.9660	0.1585	approx. normal
Shapiro-Wilk (virginica)	W = 0.9622	0.1098	approx. normal
Levene's test (equal variances)	stat = 19.72	2.59×10^{-8}	variances differ

All three groups pass the normality check individually, but Levene's test shows the assumption of equal variances across groups is violated (larger species have more variable petal lengths). Section 13.0 therefore also reports Welch's ANOVA, which does not assume equal variances, and the non-parametric Kruskal-Wallis test, which assumes neither normality nor equal variances.

13.0 ANOVA Results

Classical one-way ANOVA: $F(2, 147) = 1179.03$, $p = 3.05 \times 10^{-91}$ — reject H0. Eta-squared = 0.941, meaning species membership explains about 94% of the variance in petal length -- an extremely large effect.

Welch's ANOVA (robust to unequal variances): $F(2, 78.05) = 1826.58$, $p = 2.85 \times 10^{-66}$ — same conclusion.

Kruskal-Wallis (non-parametric, robust to both normality and variance violations): $H = 130.41$, $p = 4.80 \times 10^{-29}$ — same conclusion.

All three tests agree: mean petal length differs significantly across the three species.

Tukey HSD post-hoc (pairwise comparisons):

Group 1	Group 2	Mean Diff.	p (adj.)	95% CI	Reject H0?
setosa	versicolor	2.796	< 0.001	(2.592, 3.000)	Yes
setosa	virginica	4.088	< 0.001	(3.884, 4.292)	Yes
versicolor	virginica	1.292	< 0.001	(1.088, 1.496)	Yes

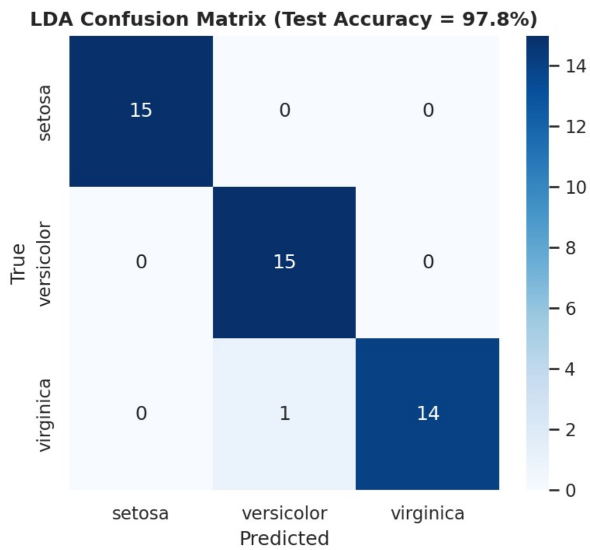
Every pairwise comparison is significant: all three species have distinct mean petal lengths.

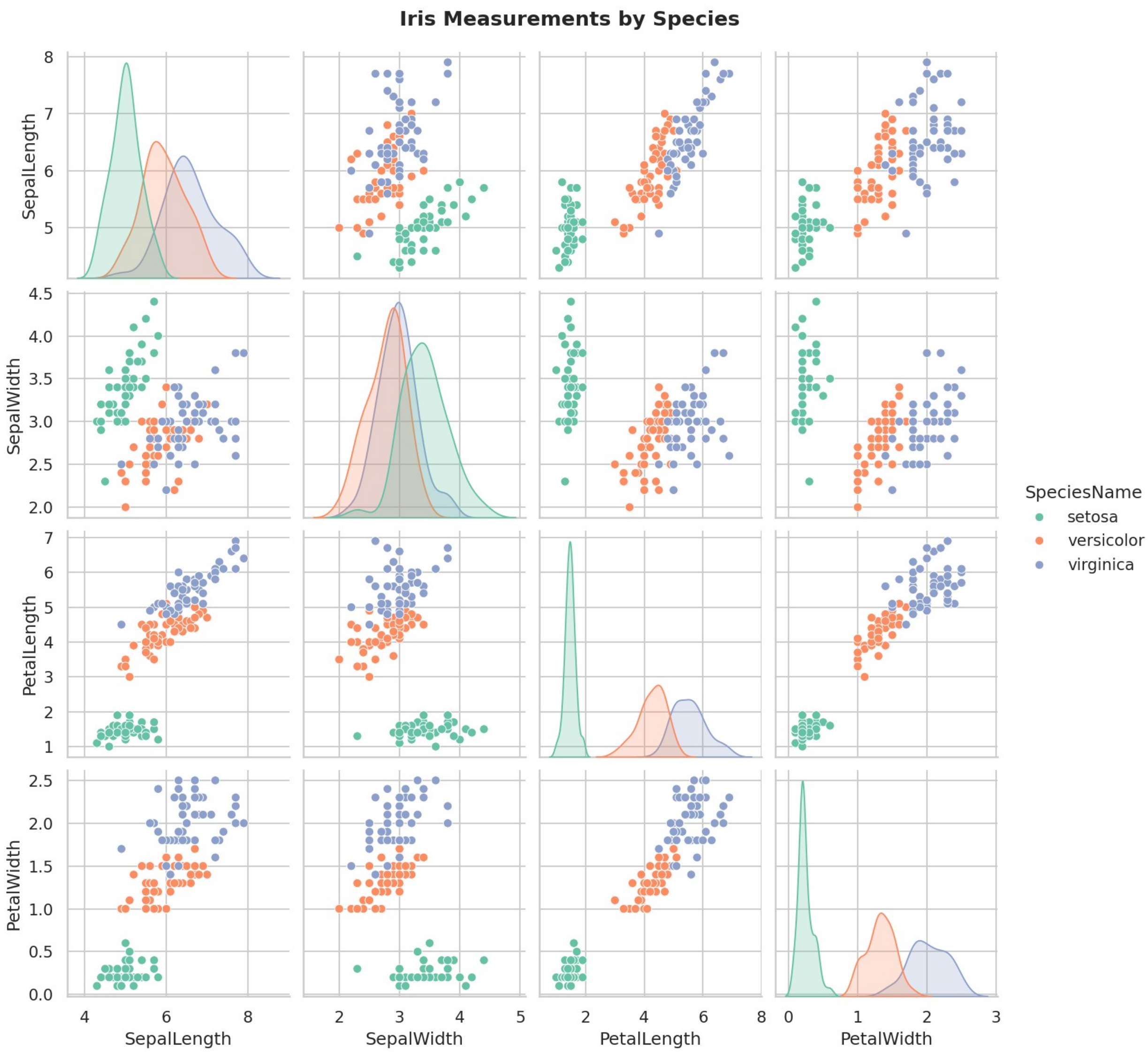
14.0 Extension: LDA Classification

A Linear Discriminant Analysis (LDA) classifier was trained on all four measurements (Sepal Length, Sepal Width, Petal Length, Petal Width) using a stratified 70/30 train/test split (105 training, 45 test observations, seed = 42).

Species	Precision	Recall	F1-score	Support
setosa	1.00	1.00	1.00	15
versicolor	0.94	1.00	0.97	15
virginica	1.00	0.93	0.97	15

Overall test accuracy: 97.8% (44/45 correct) — a single virginica flower was misclassified as versicolor, consistent with the small overlap visible between those two species in the pairplot below.


LDA Confusion Matrix (Test Accuracy = 97.8%)
True
setosa
versicolor
virginica
15
0
0
0
15
0
0
1
14
Predicted
setosa
versicolor
virginica
14
12
10
8
6
4
2
0



15.0 Conclusion

Mean petal length differs significantly across all three iris species (classical ANOVA, Welch's ANOVA, and Kruskal-Wallis all agree, with an extremely large effect size of $\eta^2 = 0.941$), and Tukey's post-hoc test confirms every pairwise difference is significant. This strong separation is also reflected in the LDA classifier, which predicts species from the four measurements with 97.8% test accuracy, misclassifying only a single borderline versicolor/virginica flower.

16.0 Python Code (Part B)

```
"""
MATH F432 Applied Statistical Methods -- Assignment 2, Part B
One-Way ANOVA + Classification on the Iris dataset (iris.csv).

Tasks:
(a) One-way ANOVA: does mean Petal Length differ across the 3 species
    (setosa, versicolor, virginica)? H0: mu1 = mu2 = mu3, Ha: at least one differs.
(b) Check ANOVA assumptions: normality per group (Shapiro-Wilk) and
    homogeneity of variance (Levene's test); if variances are unequal, also
```

```
    run Welch's ANOVA and the non-parametric Kruskal-Wallis test as
    robustness checks, and Tukey's HSD as the post-hoc test.
(c) Extension: a Linear Discriminant Analysis (LDA) classifier predicting
    species from all four measurements, evaluated on a held-out test split.
"""
import os

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pingouin as pg
from scipy import stats
from statsmodels.stats.multicomp import pairwise_tukeyhsd
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

sns.set(style="whitegrid")

BASE_DIR = os.path.dirname(os.path.abspath(__file__))
FILE_PATH = os.path.join(BASE_DIR, "iris.csv")
FIG_BOX = os.path.join(BASE_DIR, "fig_petal_length_by_species.png")
FIG_PAIR = os.path.join(BASE_DIR, "fig_iris_pairplot.png")
FIG_CM = os.path.join(BASE_DIR, "fig_lda_confusion_matrix.png")

ALPHA = 0.05
SEED = 42
SPECIES_MAP = {0: "setosa", 1: "versicolor", 2: "virginica"}
FEATURES = ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth"]

try:
    df = pd.read_csv(FILE_PATH)
except FileNotFoundError:
    print(f"Error: File not found at '{FILE_PATH}'")
    raise SystemExit(1)

df["SpeciesName"] = df["Species"].map(SPECIES_MAP)

print("--- Descriptive Statistics: Petal Length by Species ---")
print(df.groupby("SpeciesName")["PetalLength"].describe().round(3))

groups = [g["PetalLength"].values for _, g in df.groupby("SpeciesName")]
species_order = sorted(df["SpeciesName"].unique(), key=lambda s: list(SPECIES_MAP.values()).index(s))

# ------------------------------------------------------------------------
# Assumption checks
# ------------------------------------------------------------------------
print("\n--- Assumption Checks ---")
for name, g in zip(species_order, groups):
    sh_stat, sh_p = stats.shapiro(g)
    print(f"  Shapiro-Wilk ({name}): W={sh_stat:.4f}, p={sh_p:.4f} "
          f"-> {'not normal' if sh_p < ALPHA else 'approx. normal'}")

lev_stat, lev_p = stats.levene(*groups)
print(f"  Levene's test (equal variances): stat={lev_stat:.4f}, p={lev_p:.3e} "
      f"-> {'variances differ' if lev_p < ALPHA else 'variances equal'} at alpha={ALPHA}")

# ------------------------------------------------------------------------
# One-way ANOVA
# ------------------------------------------------------------------------
print("\n--- One-Way ANOVA: PetalLength ~ Species ---")
f_stat, anova_p = stats.f_oneway(*groups)
grand_mean = df["PetalLength"].mean()
ss_between = sum(len(g) * (g.mean() - grand_mean) ** 2 for g in groups)
```

```
ss_total = ((df["PetalLength"] - grand_mean) ** 2).sum()
eta_sq = ss_between / ss_total
print(f"  F({len(groups)-1}, {len(df)-len(groups)}) = {f_stat:.2f}, p = {anova_p:.3e}")
print(f"  Eta-squared (effect size) = {eta_sq:.4f}")
print(f"  Decision: {'Reject' if anova_p < ALPHA else 'Fail to reject'} H0 at alpha={ALPHA}")

print("\n--- Robustness Checks (variances were unequal per Levene's test) ---")
welch = pg.welch_anova(data=df, dv="PetalLength", between="SpeciesName")
print(welch.to_string(index=False))
kw_stat, kw_p = stats.kruskal(*groups)
print(f"  Kruskal-Wallis (non-parametric): H={kw_stat:.2f}, p={kw_p:.3e}")

print("\n--- Tukey HSD Post-hoc Test ---")
tukey = pairwise_tukeyhsd(df["PetalLength"], df["SpeciesName"], alpha=ALPHA)
print(tukey)

# ---------------------------------------------------------------------------
# Extension: LDA classification
# ---------------------------------------------------------------------------
print("\n--- Extension: LDA Classification (Species ~ all 4 measurements) ---")
X = df[FEATURES]
y = df["SpeciesName"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=SEED, stratify=y
)
lda = LinearDiscriminantAnalysis()
lda.fit(X_train, y_train)
y_pred = lda.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print(f"  Train/test split: {len(X_train)}/{len(X_test)} (stratified, seed={SEED})")
print(f"  Test accuracy: {acc:.4f}")
print(classification_report(y_test, y_pred))
cm = confusion_matrix(y_test, y_pred, labels=species_order)
print("Confusion matrix (rows=true, cols=predicted):")
print(pd.DataFrame(cm, index=species_order, columns=species_order))

# ---------------------------------------------------------------------------
# Figures
# ---------------------------------------------------------------------------
plt.figure(figsize=(7, 5))
sns.boxplot(data=df, x="SpeciesName", y="PetalLength", order=species_order,
            hue="SpeciesName", palette="Set2", legend=False)
sns.stripplot(data=df, x="SpeciesName", y="PetalLength", order=species_order,
              color="black", alpha=0.35, size=3)
plt.title("Petal Length by Species", fontsize=14, fontweight="bold")
plt.xlabel("Species")
plt.ylabel("Petal Length (cm)")
plt.tight_layout()
plt.savefig(FIG_BOX, dpi=150)
plt.close()

g = sns.pairplot(df, vars=FEATURES, hue="SpeciesName", palette="Set2", diag_kind="kde")
g.fig.suptitle("Iris Measurements by Species", y=1.02, fontsize=14, fontweight="bold")
g.savefig(FIG_PAIR, dpi=150)
plt.close()

plt.figure(figsize=(5.5, 5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=species_order, yticklabels=species_order)
plt.title(f"LDA Confusion Matrix (Test Accuracy = {acc:.1%})", fontsize=12, fontweight="bold")
plt.xlabel("Predicted")
plt.ylabel("True")
plt.tight_layout()
plt.savefig(FIG_CM, dpi=150)
plt.close()
```

```
print(f"\nFigures saved: {FIG_BOX}, {FIG_PAIR}, {FIG_CM}")
```